GGG-S-121E
AMENDMENT-1
March 25, 1976

AMENDMENT

TO

FEDERAL SPECIFICATION

SCREWDRIVER AND SCREW STARTER, HAND

This amendment, which forms a part of Federal Specification GGG-S-121E, dated June 10, 1974, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 1

Paragraph 1.2.1, under type I, class 5: Change design B to read "Design B - Bolster, forged."

PAGE 3

Paragraph 3.4: Add "When specified (see 6.1), blades shall be covered with an insulating material for the purpose of preventing short circuiting of electrical apparatus while adjustments or maintenance is being performed. Insulation covering shall meet the dielectric strength test in 4.4.12."

Paragraph 3.5: At the end of the first sentence, add "and 4.4.7". On the eighth line, delete "size" and substitute "side".

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Paragraph 3.7: After third sentence, add new sentences as follows: "The cushion grip shall not project past the butt end of the handle, and both ends of the cushion grip shall blend smoothly with the handle. The cushion grip shall be ribbed or fluted on the exterior surface to prevent slippage of the hand."

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Table I, seventh column: Change bending moment requirement for 9/64 inch blade tip width sizes from "60" inch-pounds to "40" inch-pounds; and in fifth column, change handle length dimension for 7/32 inch blade tip width screwdrivers from "3-1/2" to "3-3/8".

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Table II, fifth column: Change "Min." in heading to "+1/2, -0".

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Paragraph 3.8.4., fourth and fifth sentences: Delete and substitute, "Design B blades shall be of the bolster type. Blades over 6 inches long shall have bolsters with wrench flats, or be of square or hexagonal cross-section. Blades having a length of 6 inches or less may be with either a plain round bolster or with wrenching facilities. The outside diameter or width across flats dimension of bolsters shall be at least 50 percent larger than the diameter of the blade. When the blade and handle are assembled, the bolster shall fit tightly against the handle."

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PAGE 7 (CON.)

Table IV: Add 8 inches long by 1/4 inch tip width size after the 4 inches long by 1/4 inch tip width size (same data in all other columns). Also within the bracket for the 3/8 tip width sizes, add 6 and 8 inch long blade lengths, and change 4-5/8 length of handle to 4-3/8. Delete the 6 inch long, 3/16 tip width screwdriver as listed, and add at top of the list in columns 1 through 8, add the following: "6", "3/16", "0.025", "15/16", "3-1/4", "25", "100", and "D".

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Table V, fourth column: Change diameter or distance across corners for 5/32 and 5/16 inch tip widths from "5/8" and "1-1/8" to "9/16" and "1-1/16", respectively.

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Paragraph 3.8.7, fourth sentence: Delete "3/16-inch" and substitute "1/8-inch".

Table VII, fourth column: Change "min." in heading to "+1/8, -1/16", and change diameter or distance across corners from "7/16" and "1/2" to "13/32" and "7/16", respectively. Also, after the 2 inch blade length for the 3/32 inch tip width, add "3"; and within the bracket for the 1/8 inch tip width sizes, add 1/2 and 3/4 inch long blade lengths.

Paragraphs 3.9, 3.9.1, 3.9.2, 3.9.3: Delete and substitute:

"3.9 Type III, screw starters, hand. Type III screw starters shall be for use on straight-slotted screws, and shall consist essentially of a screw-slot gripping mechanism and a handle.

3.9.1 Screw-slot gripping mechanism. The screw-slot gripping mechanism shall grip the screw internally within the slot of the screw head by a wedge action between the blade and the screw slot, and shall be suitable for starting or removing screws from restricted places. The tip and the part of the gripper which engages the screw slot shall have a Rockwell C45-58 hardness.

3.9.2 Handle. The handle shall be the portion of the screw starter that is held in the hand, and shall be made of either plastic or metal of a suitable size. Screwdriver type handles shall be in accordance with 3.5. Metal handles shall be knurled.

3.9.3 Dimensions. Type III screw starters shall conform to the dimensions in table VIII for the size specified (see 6.1)."

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Table VIII: Delete and substitute:

"TABLE VIII. Type III screw starter, hand

Tip width at end +1/16, -1/64	Thickness of tip at end ±0.004	Overall length (Minimum)
Inch	Inch	Inches
1/8	0.017	6-3/4
3/16	.030	3-5/8
9/32	.032	5-1/8
1/4	.037	7-1/4
1/4	.037	9-1/4
9/32	.032	9-1/4
3/16	.030	11-1/4

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Paragraph 3.10.3.1, fifth sentence: Delete.

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Table XI: Delete columns 6 and 7 for diameter of shank.

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Table XII: Delete fourth column for diameter of shank.

Paragraph 3.11.2, first sentence: Delete "three flat tip bits (see 3.11.2.1)" and substitute: "one flat tip and one Phillips bit (see 3.11.2.1 and 3.11.2.2)".

PAGE 16

Paragraph 3.11.2.1: Delete and substitute:

"3.11.2.1 Flat tip bits. Flat tip bits for type V, class 2 screwdrivers shall conform to 3.4, 3.4.1, 3.4.2, figure 15 and tables XV and XVI. One bit number 2, 5, and 8 shall be furnished with each style 1, 2, 3 screwdrivers, respectively. When specified (see 6.1), bits shall be supplied separately."

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Paragraph 3.11.2.2: Delete and substitute:

"3.11.2.2 Phillips cross-point bits. Phillips cross-point bits for type V, class 2 screwdrivers shall conform to applicable requirements of 3.10.3, 3.10.3.1, figure 15, and tables XV and XVII. One bit number 2, 4, and 6 shall be furnished with each style 1, 2, and 3 screwdrivers, respectively. When specified (see 6.1), bits shall be supplied separately."

Paragraph 3.11.2.3: Add to the end of the first sentence: "in lieu of the flat tip and the Phillips tip bits."

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Table XIX: Delete and substitute:

Blade			Handle		Torsional Moment (Min.)
Exposed Length ± 1/4	Point Size	Diameter ± 0.010	Diameter (Min.)	Length (Min.)	
Inch	Number	Inch	Inches	Inches	Inch-Pounds
1-1/4	0	1/8	7/16	1-1/2	6
2-1/2	0	1/8	5/8	2-1/4	6
4	0	1/8	7/16	1-1/2	6
1 1/	1	3/16	13/16	1-1/2	25
3	1	3/16	13/16	3-1/4	25
4	1	3/16	13/16	3-1/4	25
8 2/	1	3/16	13/16	3-1/4	25
1-1/2 1/	2	1/4	1	1-1/2	40
4	2	1/4	1	3-1/2	40
7	2	1/4	1	3-1/2	40
8 2/	2	1/4	1	3-1/2	40
10	2	1/4	1	3-1/2	40
6	3	5/16	1	3-7/8	90
8	4	3/8	1-1/8	4-5/8	150

1/ Close quarter screwdrivers.
2/ Extra long blade."

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Figure 24: Change handle length from 2 inches, plus 3/16, minus 1/16, to "2 inches min."

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Table XXV: Add the following sizes:

Clutch size	Blade length ± 1/4	Handle dimensions		Torsional moment (minimum)
		*Diameter or distance across flats (min.)	Length (min.)	
Inch	Inches	Inch	Inches	Inch-pounds
5/32	2	13/16	2-5/8	30
3/16	2	13/16	3-1/4	40

Change footnote to read "*When handles with inserts are furnished, the diameter of the handles shall not be less than 1-1/32".

Figure 27: Change included angle from $20^{\circ} \pm 3^{\circ}$ to "145° min., 155° Maximum".

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Table XXVI: Change R_2 dimensions for 1/4-inch clutch size from ".0768 min., .0788 max." to ".0780 min., .0800 max."

Table XXVII, on last line and in last box, change sampling data reference for examination of preparation for delivery from "4.4.12" to "4.5".

PAGE 31

Paragraph 4.4.3, fourth and fifth line: Delete the sentence beginning "The load on the point end ...".

PAGE 32

Paragraph 4.4.5, first line after the word "vise", insert "or milled slot".

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Add paragraph 4.4.12 as follows:

"4.4.12 Dielectric strength. When applicable, two sample units shall be tested for dielectric strength. One sample shall be tested after immersion in transformer oil for a period of 5 hours, and one sample shall be tested after immersion in water for a period of 5 hours. After the insulation has been wiped dry, it shall be wrapped in metal foil to within 7/8 inch of the nearest exposed metal part. An electrical lead shall be connected to the exposed tip, and another lead to the foil wrapping, and then subjected to a voltage of not less than 2200 volts, 60 hertz, a.c. After a period of one minute, acceptable insulation coverings shall show no signs of breakdown."

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